Experiment No. :- 06

1] Make A Folder on Local Machine

2] Access that folder on VSCode
Example:
ai-docker-app(Folder name is user defined)
Open this folder in VS Code (or any editor).

3] Create app.py (Make a file in VsCode name with extension .py)

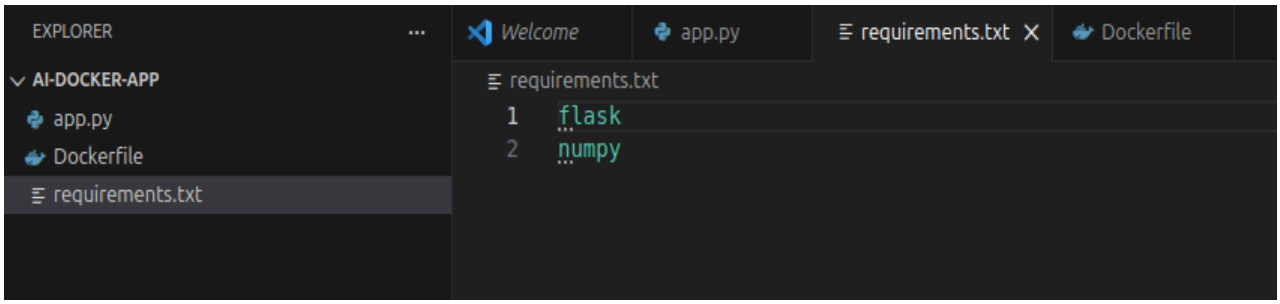
eg.

```
from flask import Flask, request, jsonify

app = Flask(__name__)

@app.route('/')
def home():
return "AI Model is Running!"

@app.route('/predict', methods=['POST'])
def predict():
data = request.json['number']
result = data * 2
return jsonify({'result': result})

if __name__ == '__main__':
app.run(host='0.0.0.0', port=5000)
```

4] Create requirement.txt (inside the VSCode in same folder)

add this info

flask
numpy

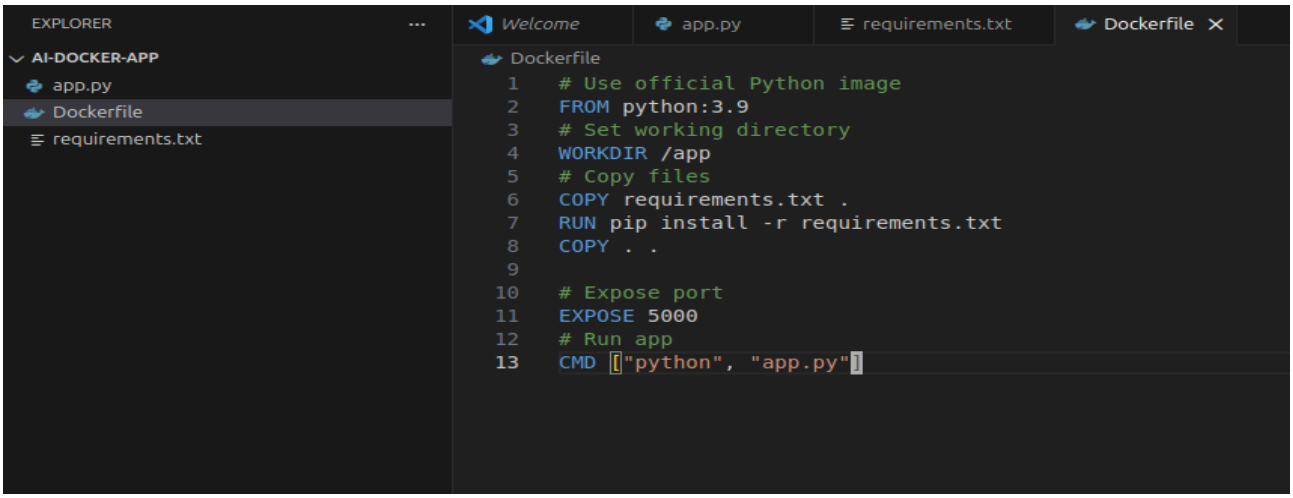


5] Create Dockerfile (inside the VSCode in same folder)

add this info in Dockerfile

```
# Use official Python image
FROM python:3.9
# Set working directory
WORKDIR /app
# Copy files
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .

# Expose port
EXPOSE 5000
# Run app
CMD ["python", "app.py"]
```

eg :-

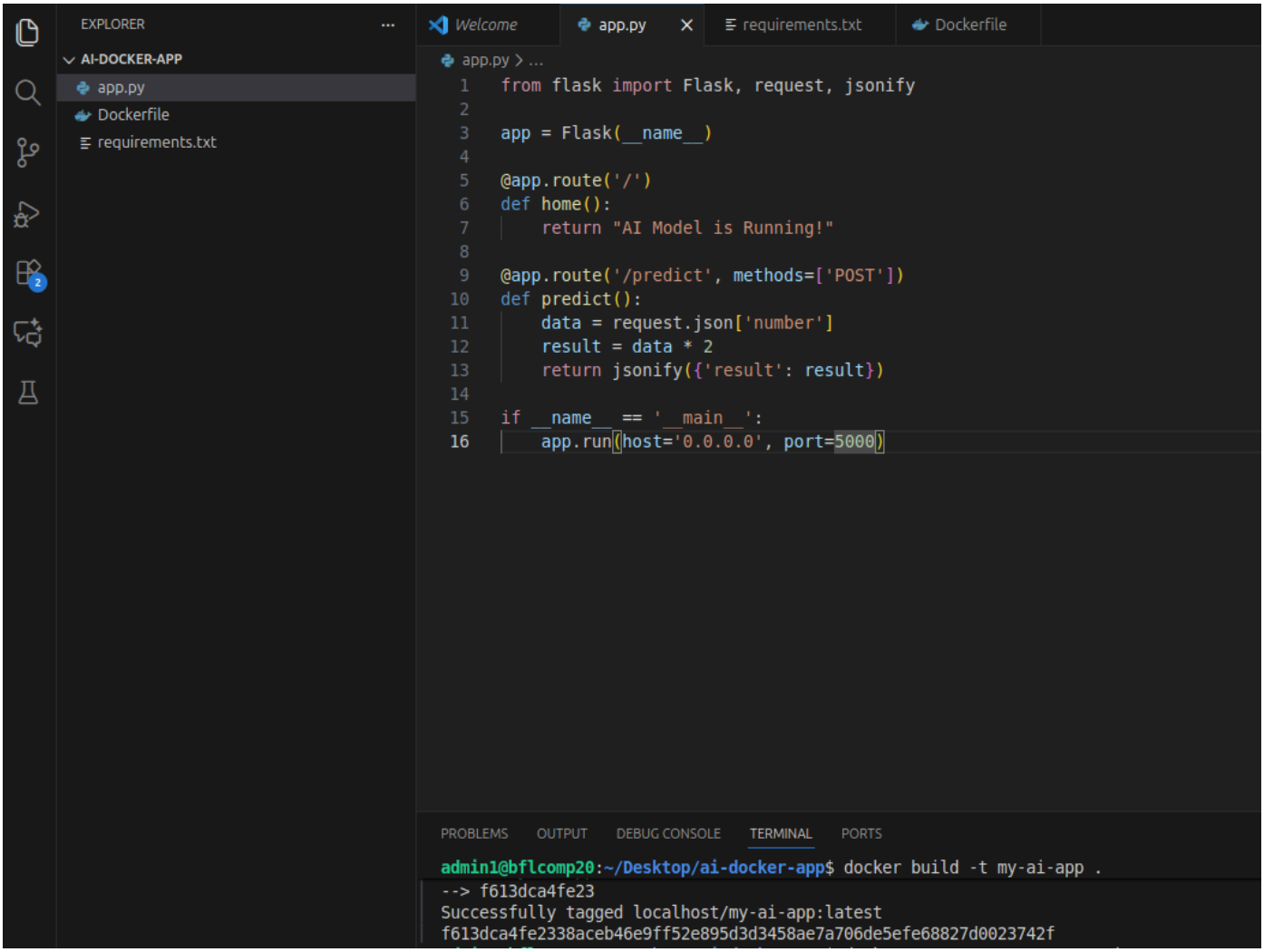


6] Build Docker Image

Open terminal inside your project folder:
Run:

docker build -t my-ai-app .

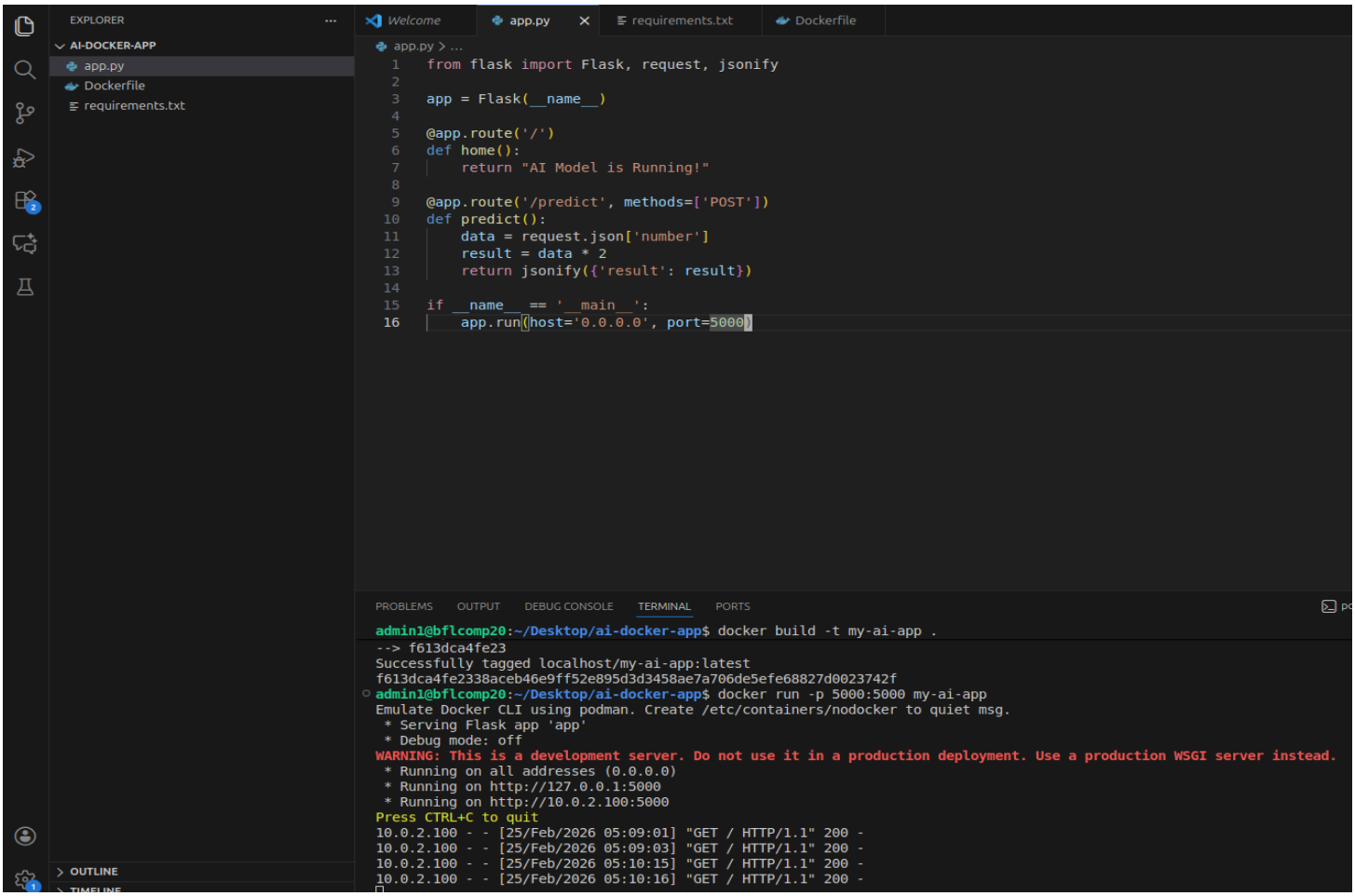
eg.



7] Run Docker Container

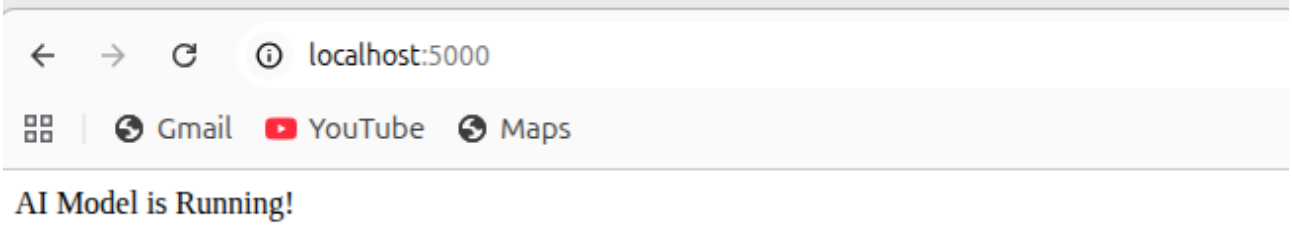
Run this:
docker run -p 5000:5000 my-ai-app

eg .



8] Test It On Local Machine

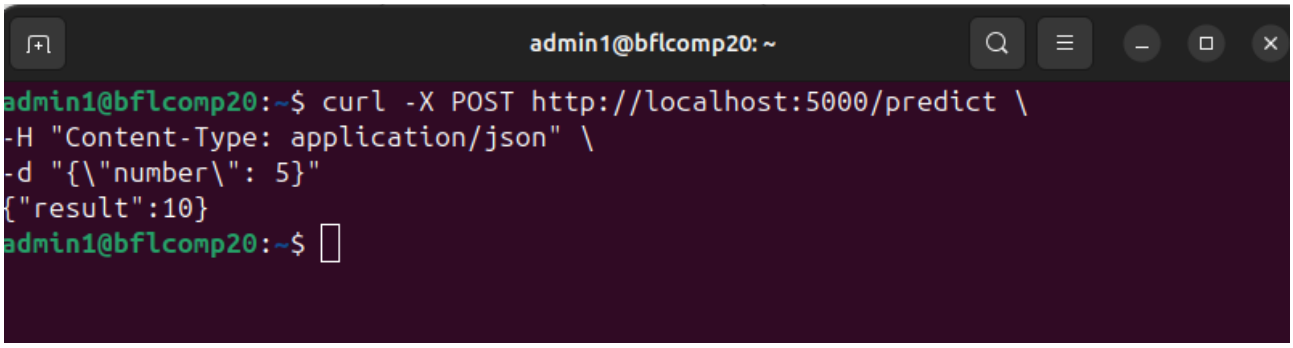
Open browser and go to:
http://localhost:5000
You should see:
AI Model is Running!
Eg .



9] Test Prediction API

Open another terminal and run:

```
curl -X POST http://localhost:5000/predict \
-H "Content-Type: application/json" \
-d "{\"number\": 5}"
```

You should see: (OUTPUT)

{"result":10}